

46-937: Individual Prediction Directions and Rubric

Due 11:59 PM (Eastern Time), FRIDAY, Sept 25, 2026.

The following details what you will be submitting for the individual prediction portion of the final project. You will be turning in the test data prediction files using Google forms (provided on Canvas), while your code (and eventually the final reports) will be submitted through Gradescope as PDF files. If you have any questions about any of the information below, please post on Piazza.

1. *Training data.* You will use the training dataset (train.csv) located in the project folder on Canvas to build one or more models to predict the following **9 response variables**:

- Q0_TOTAL_ASSETS
- Q0_TOTAL_LIABILITIES
- Q0_TOTAL_STOCKHOLDERS_EQUITY
- Q0_GROSS_PROFIT
- Q0_COST_OF_REVENUES
- Q0_REVENUES
- Q0_OPERATING_INCOME
- Q0_OPERATING_EXPENSES
- Q0_EBITDA

I originally gathered the data [from a Kaggle contest](#), but **I used it to create synthetic data for this project**. Besides the 9 response variables, there are lagged versions of these variables that you can use as predictors along with other variables - all of which are fully explained in the data dictionary file on Canvas (named data_dictionary.txt). The only column that you should NOT use as a predictor variable is the 'Id' column. This is NOT a clean dataset, and you will need to think of proper ways for handling missing values.

2. *Training models.* In terms of which model or models you can use to generate predictions for the nine response variables, I am giving you the freedom to **turn in predictions via any technique that you deem appropriate**. However, when it comes to turning in the final report (rubric to be provided soon) **you will need to include a model comparison (with some form of cross-validation) between 3 types of models**:

- Naive baseline model - just using the average of each response variable as the prediction, ignoring all other predictors
- Not deep learning model - any type of model or models that are NOT a form of deep learning that we considered in this class
- Deep learning model - some type of deep learning model(s) covered in this class!

Your individual predictions do NOT have to be from a deep learning model, but I will expect you to compare the performance of one in your individual written portion of the final report.

3. *Submitting predictions.* Using your trained model(s), you will submit predictions for each of the 9 response variables based on the test dataset located on Canvas named test_individual.csv. Using [this Google form](#), **you will upload a csv file** that contains your predictions for each observation in the test dataset with 10 columns: the Id column and then a column for each of the 9 response variables listed above. Your column names need to match the response variable names in order to receive credit. **Your prediction submissions are due Friday Sept 25th by 11:59 PM ET.** Additionally, you will need to **submit your code on Gradescope as a PDF by the same submission deadline.** It does NOT need to be fully documented code, and it does NOT need to include a full model comparison that will be required for the report. I just need proof of code demonstrating how you fit the model and then used it to generate predictions (as detailed in the grading rubric section below).
4. *Evaluation criteria.* Since you will be submitting predictions for 9 response variables, **the evaluation metric will be the average symmetric mean absolute percentage error (sMAPE) computed on the test data across the 9 response variables.** In other words, I will compute the test sMAPE for each of the 9 response variables separately, and then will take the average of the 9 values as the overall assessment of your submission. To be explicit, for a single target variable m the sMAPE across n observations is defined as:

$$\text{sMAPE}_m = \frac{100\%}{n} \sum_{i=1}^n \frac{|y_{i,m} - \hat{y}_{i,m}|}{\frac{1}{2}(|y_{i,m}| + |\hat{y}_{i,m}|)} \quad (1)$$

For $M = 9$ target variables, the overall evaluation metric is the average sMAPE across all targets:

$$\overline{\text{sMAPE}} = \frac{1}{M} \sum_{m=1}^M \text{sMAPE}_m = \frac{1}{M} \sum_{m=1}^M \left(\frac{100\%}{n} \sum_{i=1}^n \frac{|y_{i,m} - \hat{y}_{i,m}|}{\frac{1}{2}(|y_{i,m}| + |\hat{y}_{i,m}|)} \right) \quad (2)$$

5. *Grading rubric.* The grade for your individual predictions will be based on a comparison with two other models:

- Baseline naive model - as described earlier, just taking the average on the training data without using any predictors
- AI - I used AI to create a model (or models - I'm not telling you!) for this dataset with a single prompt, but I will not tell you any more than that.

Your prediction grade (out of 10 points) will be computed based on the average sMAPE of your model, the baseline model, and my AI predictions:

- 10/10: your average sMAPE $<$ my AI sMAPE
- 8/10: Your average sMAPE is between a 30% to 45% reduction in sMAPE compared to the baseline (this is percentage reduction relative to the baseline sMAPE, not in terms of the sMAPE scale!)
- 6/10: Your average sMAPE is between a 15% to 30% reduction in sMAPE compared to the baseline
- 4/10: Your average sMAPE is between a 1% to 15% reduction in sMAPE compared to the baseline
- 0/10: You did worse than the naive baseline. I can't help you.

This scale is subject to change: depending on the distribution of the performances displayed by your submissions. But any change I make will only improve your grade relative to this initial grading scale.